MIL-S-81018/2D(AS)
25 April 1980
SUPERSEDING
MIL-S-81018/2C(AS)
21 February 1967

MILITARY SPECIFICATION SHEET

SURVIVAL KIT CONTAINER, AIRCRAFT SEAT, WITH OXYGEN, RSSK-8D

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This detail specification sheet together with the general specification cover the requirements for a rigid seat survival kit containing oxygen and survival equipment designated as RSSK-8D.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-6117	-Terminal, Cable Assemblies, Swaged Type
MIL-H-6875	-Heat Treatment of Steels (Aircraft Practice), Process for
MIL-F-21840	-Fastener Tapes, Hook and Pile, Synthetic
MIL-S-81018	-Survival Kit Container, Aircraft Seat, with Oxygen, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, ESSD, Code 93, Lakehurst, New Jersey 08733 by using the self-addressed Standard Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:55Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-595	-Colors

MILITARY

MS20819	-Sleeve, Flared Tube Fitting
MS27983	-Fastener, Snap, Style 4
MS90389	-Cylinder, Oxygen, Nonshatterable, Welded, 1800 PSI (Straight, "U" and Spiral Types)

AIR FORCE - NAVY AERONAUTICAL

AN818	-Nut, Tube, Coupling, Short
AN833	-Elbow, Flared Tube and Bulkhead Universal, 90 Degrees

DRAWINGS

NAVAL AIR SYSTEMS COMMAND

67A261D1	-Container Lid Contour, RSSK-8A
1195AS101	-Backplate, - RSSK-8 Series
1195AS102	-Bracket, Forward Guide LH, - RSSK-8 Series
1195AS103	-Bracket, Forward Guide RH, - RSSK-8 Series
1195AS104	-Shim, Laminated, - RSSK-8 Series
1195AS105	-Shim, Laminated, - RSSK-8 Series
1195AS106	-Strap, Parachute Retention, - RSSK-8 Series Source Control Drawing
1195AS107	-Clip, - RSSK-8 Series
1195AS108	-Spacer, - RSSK-8 Series
1195AS109	-Latch, Lid, Universal, - RSSK-8 Series
1195AS113	-Bracket, Handle Protector, - RSSK-8 Series

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:55Z
Check the source to verify that this is the current version before use.

NAVAL AIR DEVELOPMENT CENTER

CC101D2	-Ejection Seat to RSSK and Torso Harness Restraint Subassembly
CL204D2	-Bracket Assembly, AN/URT-33 Beacon
CL223A3	-Elbow, Flared Tube, 5/16 to 1/4 Tubing, RSSK-8 Series
CL223D2	-Actuator Assembly, Emergency Oxygen, Manual, RSSK-8 Series
33D1341	-Hose Assembly, Oxygen, Seat Kit to Pilot and Ship Supply

(Copies of Specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply:

DOUGLAS AIRCRAFT CORPORATION

DRAWINGS

3825279	-Coordination Information, RSSK-8 Seat Attachments Points
5822131	-Seat Assembly, Rocket Catapult Personnel Ejection

(Application for copies should be addressed to the Commanding Officer, Naval Publications and Forms Center (Code 1051), 5801 Tabor Avenue, Philadelphia, PA 19120.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The RSSK-8D rigid seat survival kits shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4).

3.3 Materials and components. Materials and components shall conform to applicable specifications, standards and drawings as listed or required herein and unless modified herein to the general specification for the rigid seat survival kits, MIL-S-81018.

3.4 Design and construction. The design and construction of the RSSK-8D rigid seat survival kit shall conform to that specified in the general specification except as otherwise specified herein.

3.4.1 Operation. The operation of the RSSK-8D rigid seat survival kit shall be as specified in the general specification.

3.4.2 Kit components. The components of the RSSK-8D rigid seat survival kit shall conform to the general specification and shall consist of, but not be limited to, the following:

a. Basic container.

b. Comfort cushion.

c. Ejection seat to RSSK and torso harness restraint subassembly.

d. Release handle and container locking mechanism.

e. Container dropline assembly.

f. Emergency and bailout oxygen system.

g. Radio beacon bracket.

3.4.2.1 Basic container. The basic container of the RSSK-8D rigid seat survival kit shall be designed to conform to the requirements specified in the general specification, to the general design of figure 1 and the critical control dimensions contained in figures 4, 5, 6 and 7. The container shall have a hinge mechanism keyed for alignment on the forward edge of the thigh section. The hinge assembly shall be aluminum extrusions riveted to the extruded container lid support. The hinge assembly shall not permit separation of the lid from the container when subjected to the loads specified in the general specification or herein. The container shall have three internal locking-type latches, one each located on the right, left and aft sides of the kit.

3.4.2.1.1 Container bottom. The container bottom of the RSSK-8D rigid seat survival kit shall conform to the requirements of the general specification and to the envelope dimensions and general design of figures

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

5, 6 and 7. The container shall have two guide brackets and associated back plates as shown in figure 5 to retain the container to the seat during negative g maneuvers.

3.4.2.1.2 Contoured lid. The contoured container lid of the RSSK-8D rigid seat survival kit container shall conform to the requirements specified in the general specification, except that it shall conform to the coordinates of drawing 67A261D1 and to the envelope dimensions and general design of figures 4 and 6.

3.4.2.2 Comfort cushion. The comfort cushion of the RSSK-8D rigid seat survival kit shall conform to the requirements specified in the general specification and to the envelope dimensions and general design of figure 2.

3.4.2.3 Ejection seat to RSSK and torso harness restraint subassembly. The ejection seat to the RSSK and torso harness restraint subassembly shall conform to the requirements specified in the general specification and shall conform to the general design of figure 1.

3.4.2.4 Release handle and container locking mechanism. The release handle and container locking mechanism of the RSSK-8D rigid seat survival kit shall conform to the requirements specified in the general specification and shall conform to the general design of figure 5. The lid locking latches shall become mechanically integral with the locking mechanism in the bottom of the container when the container is locked. The mechanism shall open automatically when the release handle is pulled. The color of the handle shall be yellow approximately matching color number 33538 of FED-STD-595.

3.4.2.5 Container dropline. The container dropline assembly shall conform to the requirements specified in the general specification. The container dropline shall have one end permanently attached to the lid through footman brackets and the other end attached to the container bottom through the footman brackets.

3.4.2.6 Emergency bailout oxygen system. The emergency bailout oxygen system of the RSSK-8D rigid seat survival kit shall conform to the requirements specified in the general specification and shall be as shown in figure 3.

3.4.2.6.1 Manual actuation. Manual actuation of the oxygen system shall be accomplished by a pull on a ring located as shown in figure 3. The ring shall be permanently attached by cable to the reducer valve actuator and shall be mounted so as to maintain a horizontal position between the container lid and comfort cushion as shown in figure 1. The ring shall be a rigid configuration and shall be green in color.

Source: https://assist.dla.mil -- Downloaded: 202640-23T04:55Z
Check the source to verify that this is the current version before use.

3.4.2.7 Radio beacon bracket. The radio beacon bracket conforming to the general specification shall be installed as specified in figure 5.

3.4.3 Design load requirements. The design load requirements of the RSSK-8D rigid seat survival kit shall be as specified in the general specification, except for crash loads required by 4.8.7 of the general specification, the kit lid shall be rotated from the "z" axis forward and upward to 45 ± 2 degrees and the 40 g load applied "eyeballs down" with a time duration of 70 milliseconds.

3.4.3.1 Swaging. Swaging shall be accomplished in accordance with MIL-T-6117 and shall be carefully performed. Each swaging shall be proof tested as specified in MIL-T-6117 to 100 pounds or three times the maximum service load, whichever is greater.

3.5 Performance. The RSSK-8D rigid seat survival kit shall meet all the performance requirements of the general specification when tested as specified therein.

3.6 Size. The RSSK-8D rigid seat survival kit shall conform to the contour of the ejection seat bucket with 1/4 inch minimum clearance between the seat kit and the seat bucket around the periphery of the seat kit. Figures 4, 5, 6 and 7 provide critical control dimensions to establish proper kit to seat attachment.

3.7 Weight. The weight of the RSSK-8D rigid seat survival kit furnished under this specification, without oxygen hoses, shall be a maximum of 22-1/2 pounds.

3.8 Interchangeability. The interchangeability of the RSSK-8D rigid seat survival kit shall be as specified in the general specification and as specified herein. Unless otherwise specified, the interchangeability requirements of the RSSK-8D seat kit shall be demonstrated as specified in Section 4.

3.9 Markings. In addition to the identification and markings specified in the general specification, the kit shall contain the following decalcomania markings:

a. The markings, located under the inspection slots as shown in figure 1, shall indicate the lock and unlock positions of the container lock mechanisms and directions for manual unlock.

b. The markings, located under the handle housing as shown in figure 1, shall be a warning note to reset actuator assembly if the handle will not lock in place.

3.10 Workmanship. The workmanship shall be as specified in the general specification and herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The classification of inspection shall be as specified in the general specification.

4.3 Qualification inspection. The number of samples, tests and examinations for qualification inspection shall be as specified in the general specification and herein.

4.4 First article. The number of samples, tests and examinations for the first article inspection shall be as specified in the general specification and herein.

4.5 Quality conformance inspection. The quality conformance inspection shall be as specified in the general specification.

4.6 Quality conformance verification inspection. The quality conformance verification inspection shall be as specified in the general specification.

4.7 Test requirements. The requirements for examinations and tests shall be as specified in the general specification and herein.

4.8 Inspection methods. The inspection methods for the RSSK-8D shall be as specified in the general specification, except as otherwise noted herein.

4.8.1 Interchangeability. Three contoured container lids, from RSSK-8D seat kits submitted under contract, shall be interchanged with three container bottoms. The interchanged lid and bottom combination shall mate and lock while utilizing each of the release handles. In addition, three of the RSSK-8D seat kit automatic actuation pull cable disconnect upper halves and lower halves shall be interchanged with each other. Failure to interchange and operate as specified shall be considered unacceptable.

4.8.2 Container separation. The kit shall be suspended between the torso harness attachments and the kit to parachute straps. A

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:55Z
Check the source to verify that this is the current version before use.

load of 1500 pounds (collective) shall be applied at a rate of 3 ± 1 inches per minute and shall be maintained for a minimum of 5 seconds. There shall be no failure of the restraint parachute straps nor any container separation sufficiently to allow the container halves to separate or release the kit/parachute fittings from inside the kit. Slight distortions, warping, damage to the stitching shall be allowed. Upon completion of this test, the kit shall be subjected to and conform to the requirements of the release handle and bench actuation tests as specified in the general specification.

4.8.3 Windblast loads test. The rigid seat survival kit shall be subjected to the dynamic parameters of windblast loading specified below. The purpose of the exposure is to simulate the affects of windblast after man-seat separation. Prior to each test, the seat kit shall be securely interfaced to an appropriate parachute and torso harness. This system shall be donned on a simulated fifth percentile dummy. A partial dummy may be utilized if the torso and thigh features are provided. The windblast profile described below shall be performed once at each of the attitude conditions listed below. One seat kit shall experience all three tests. Dynamic turning shall be utilized during each windblast. Initial positions and motion are specified as follows:

WINDBLAST PROFILE

Windblast number	Initial position	Turning motion
1	Torso in vertical seat position and facing directly into airstream	Yaw 90° starboard
2	Torso prone with head facing into airstream	Yaw 90° port
3	Torso prone with feet facing into airstream	Yaw 90°

DYNAMIC PARAMETERS

Peak velocity	430 ± 40 KEAS
Rise time	80 ± 20MS
Total duration	1.5 seconds minimum

Photographic coverage of each windblast exposure test shall include a minimum of two high speed motion picture cameras. The minumum camera speed shall be 600 feet per second. Still photography and camera position shall

Source: https://assist.dla.mil -- Downloaded: 202640-23T04:55Z
Check the source to verify that this is the current version before use.

be utilized as directed by the procuring activity. Following this test, the kit shall be subjected to and conform to the requirements of the release handle, bench actuation and the high pressure oxygen leakage tests specified in the general specification.

5. PACKAGING

5.1 Preservation, packaging and packing shall be in accordance with the requirements as specified in the general specification.

6. NOTES AND CONCLUDING MATERIAL

6.1 All notes and concluding material contained in the general specification apply to this specification.

6.2 Ordering data. In addition to the ordering data contained in the general specification, the procurement document shall also specify the following:

a. Specific location of the indicator activator hole and the radio beacon bracket (see figure 5, note 4).

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing Activity:
Navy (AS)

Project Number
1660-N407

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

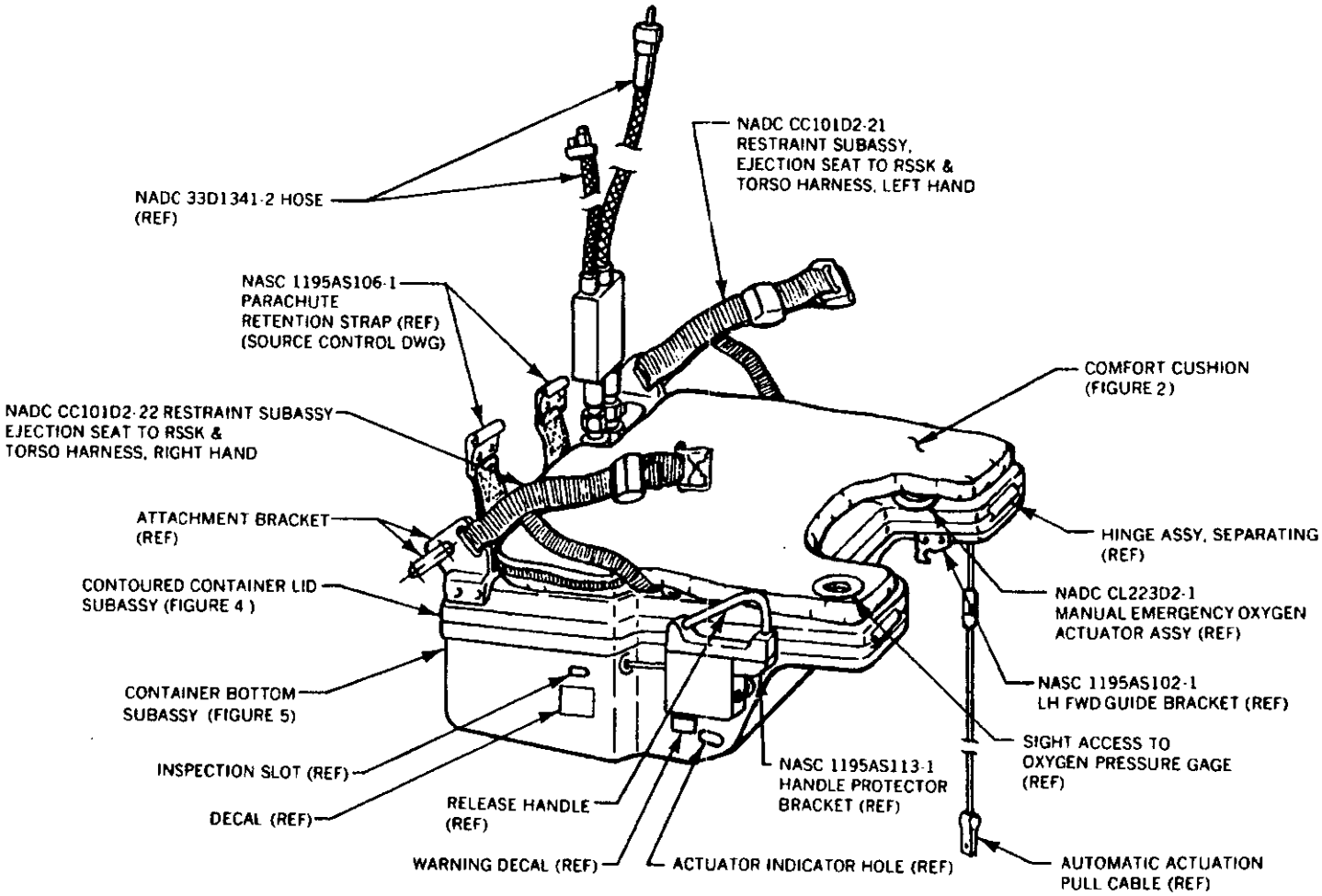


FIGURE 1. Rigid seat survival kit assembly with oxygen - RSSK-8D.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

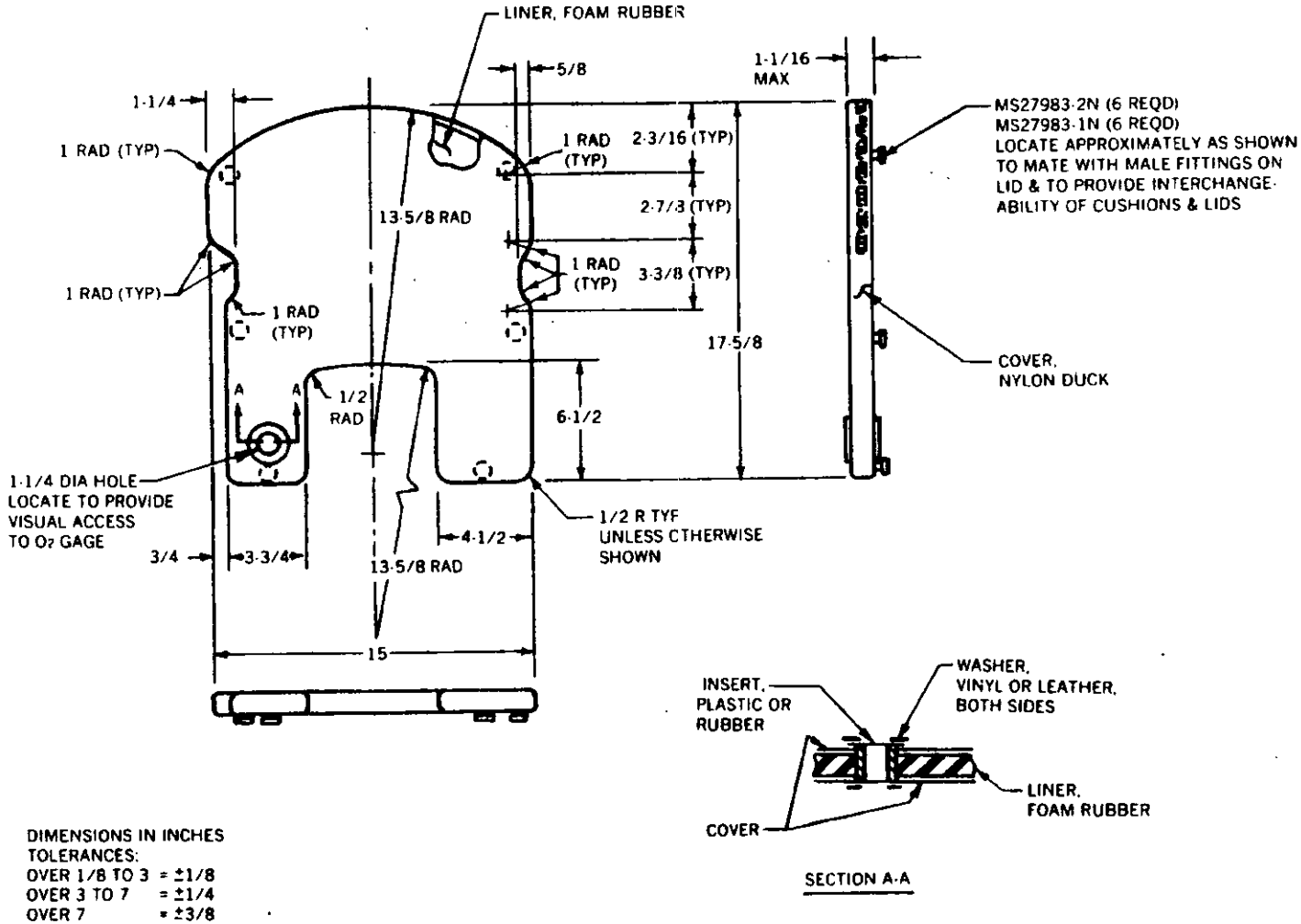


FIGURE 2. Comfort cushion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

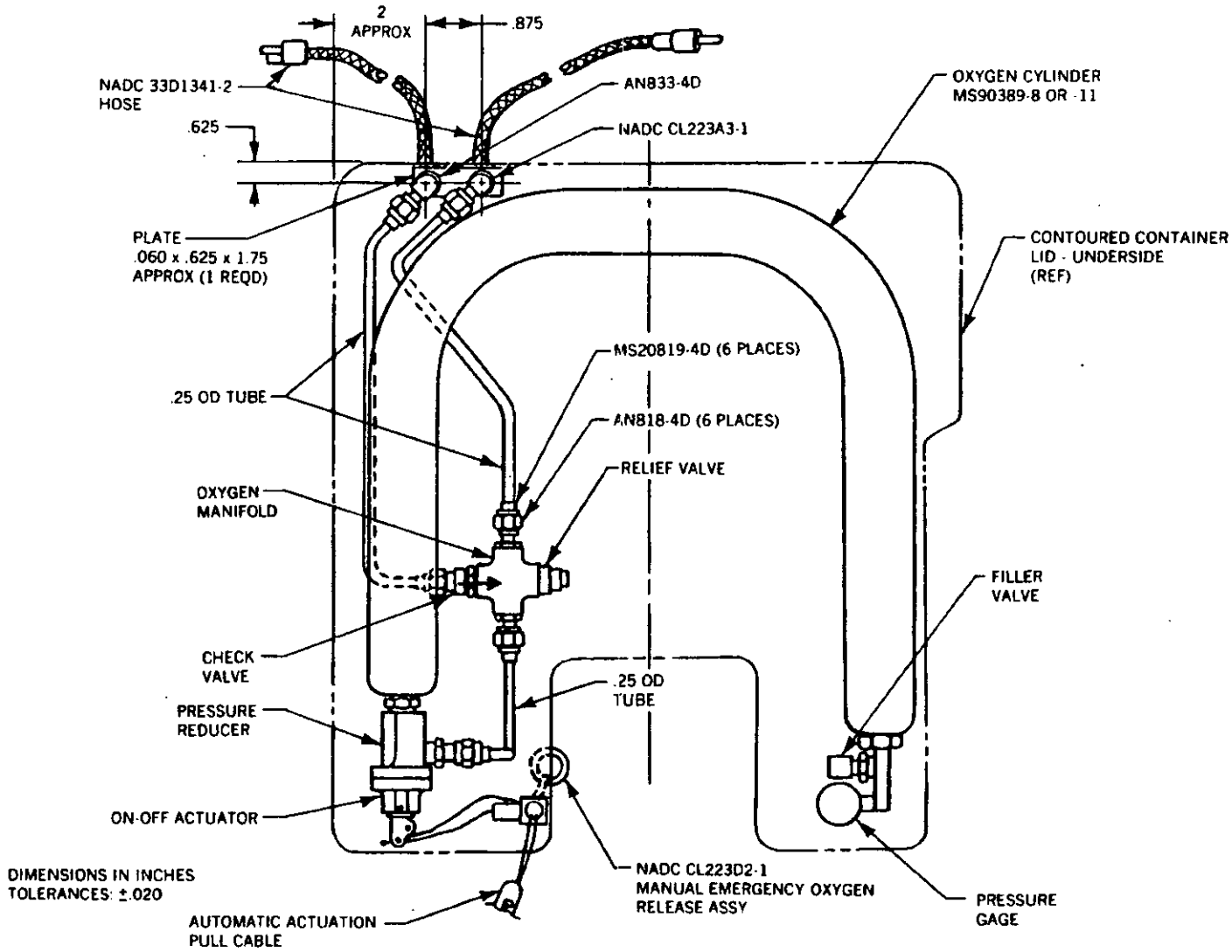


FIGURE 3. Emergency bailout oxygen system, RSSK-8D.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

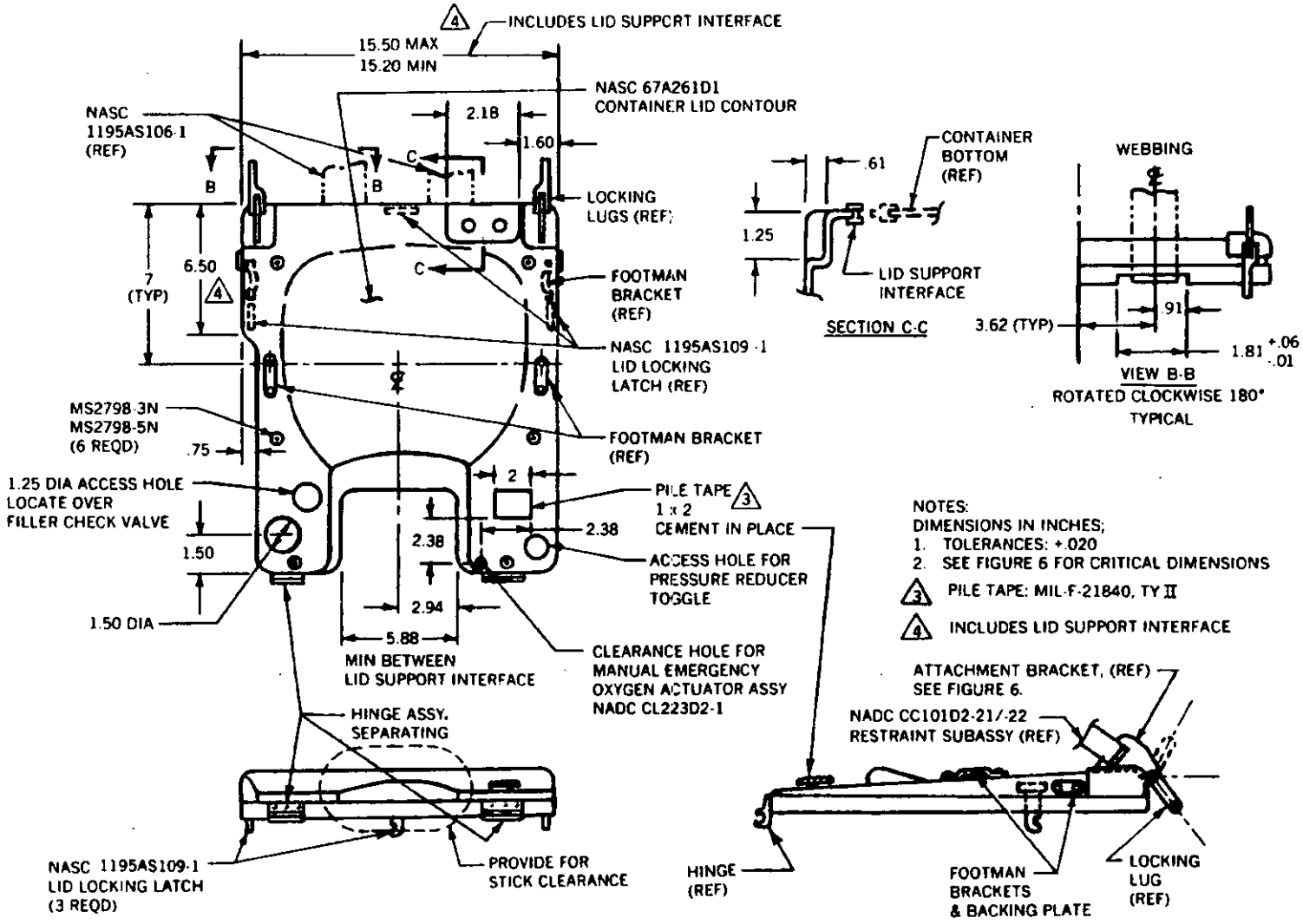


FIGURE 4. Contoured container lid subassembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

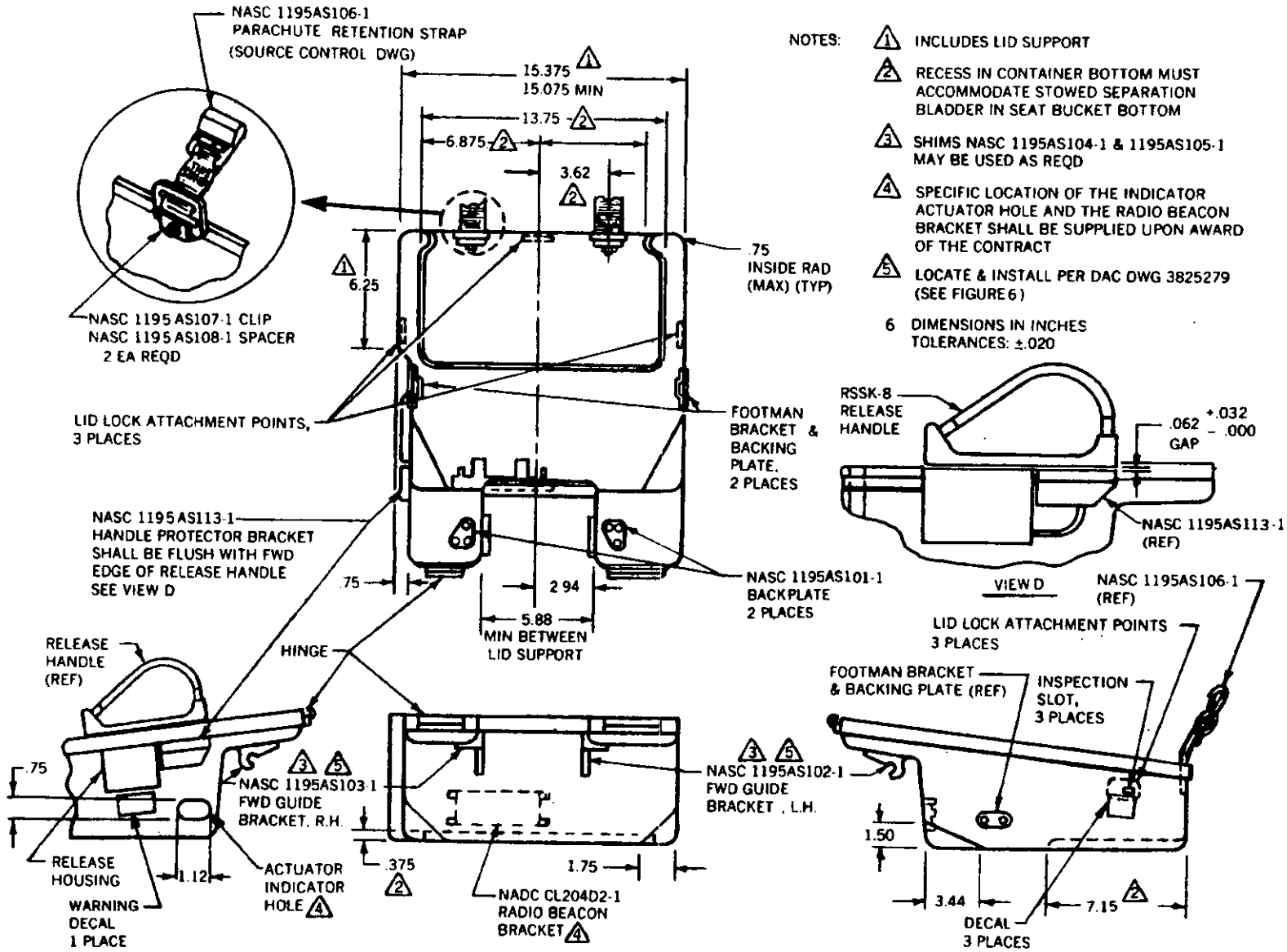
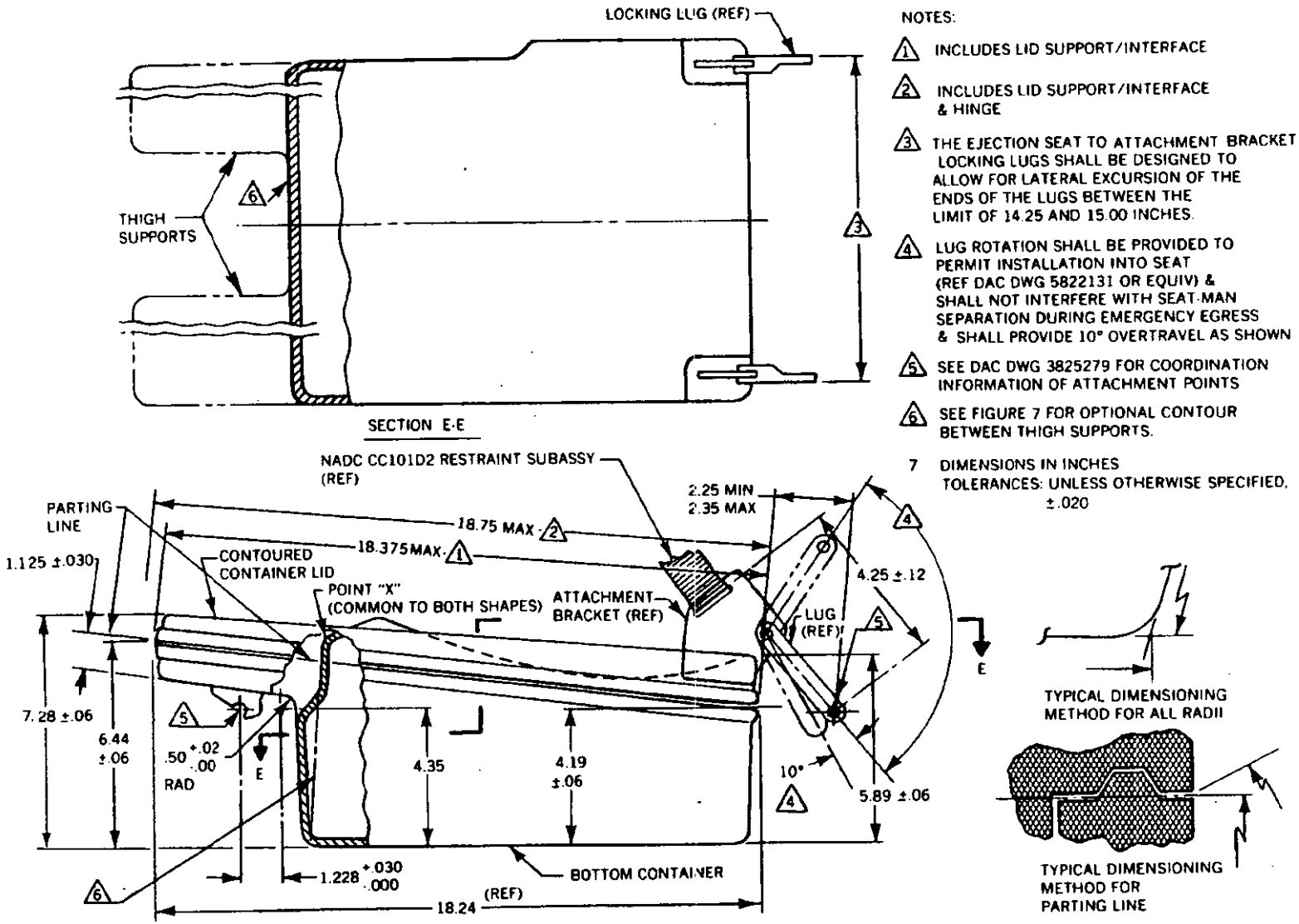


FIGURE 5. Bottom container subassembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.



NOTES:

1. INCLUDES LID SUPPORT/INTERFACE
2. INCLUDES LID SUPPORT/INTERFACE & HINGE
3. THE EJECTION SEAT TO ATTACHMENT BRACKET LOCKING LUGS SHALL BE DESIGNED TO ALLOW FOR LATERAL EXCURSION OF THE ENDS OF THE LUGS BETWEEN THE LIMIT OF 14.25 AND 15.00 INCHES.
4. LUG ROTATION SHALL BE PROVIDED TO PERMIT INSTALLATION INTO SEAT (REF DAC DWG 5822131 OR EQUIV) & SHALL NOT INTERFERE WITH SEAT-MAN SEPARATION DURING EMERGENCY EGRESS & SHALL PROVIDE 10° OVERTRAVEL AS SHOWN
5. SEE DAC DWG 3825279 FOR COORDINATION INFORMATION OF ATTACHMENT POINTS
6. SEE FIGURE 7 FOR OPTIONAL CONTOUR BETWEEN THIGH SUPPORTS.
7. DIMENSIONS IN INCHES
   TOLERANCES: UNLESS OTHERWISE SPECIFIED, ±.020

FIGURE 6. Critical dimensions (see figure 7 for optional bottom container contour). (Hinge not shown for clarity)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

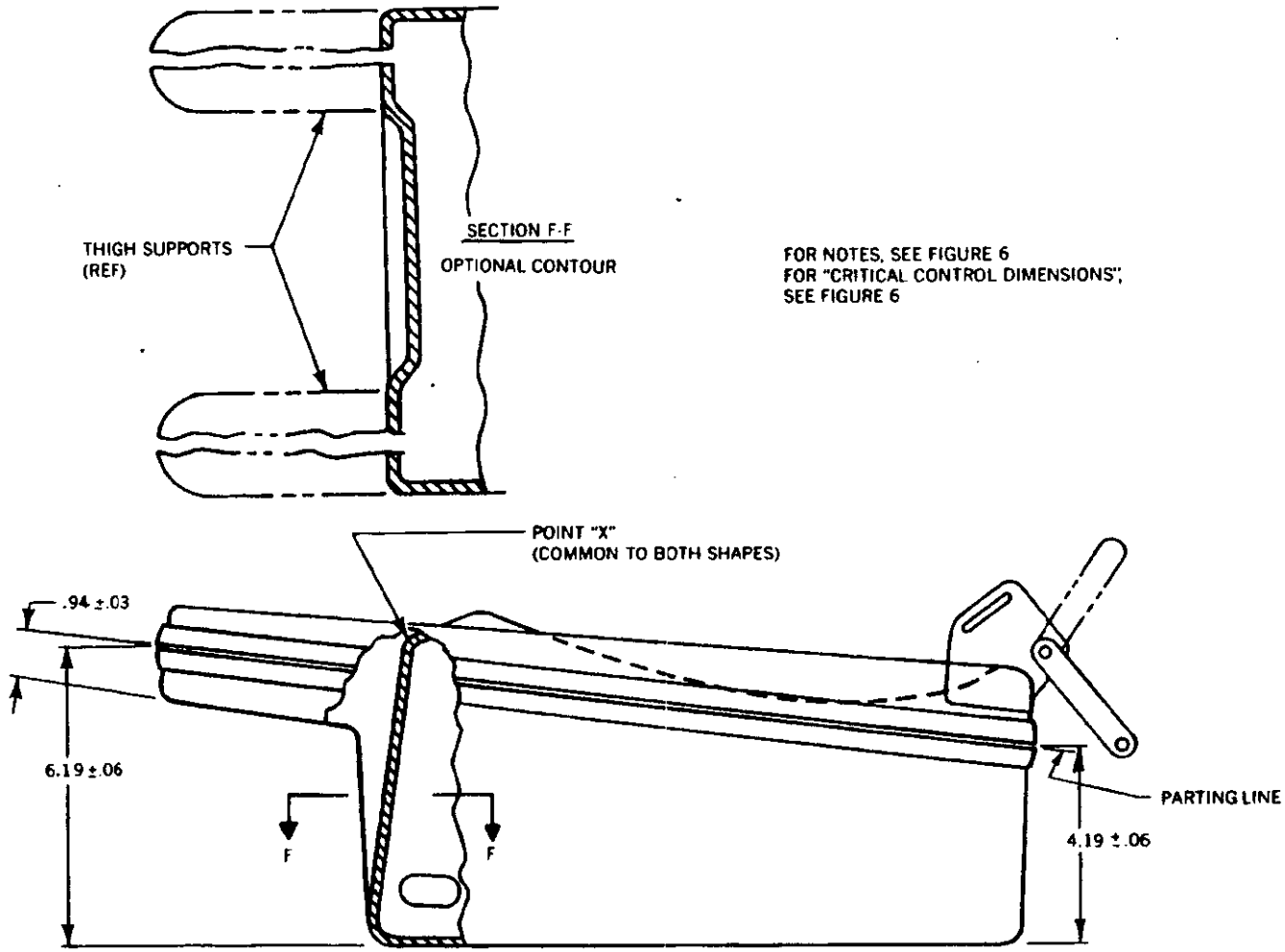


FIGURE 7. Critical dimensions for RSSK-8D, side view optional contour and dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE: MIL-S-81018/2D(AS) Survival Kit Container, Aircraft Seat, with Oxygen, RSSK-8D

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:55Z
Check the source to verify that this is the current version before use.